

Sandeep Hipparagi

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Professional Summary

Critical-thinking AI/ML Developer and Researcher seeking an Assistant Professor or Faculty position in the Department of Computer Science & Engineering / AI & Data Science. Aiming to leverage an IEEE-published research background, solid industry engineering experience and specialized expertise in agentic workflows to mentor students and bridge the gap between academic theory and cutting-edge industry implementations.

Technical Skills

- **Agentic & Resilient Architectures:** Multi-Agent Coordination, LangGraph, OpenClaw, Ollama, Self-Correction & Self-Healing Runtimes, State Management, Human-in-the-Loop (HITL) Pipelines, Retry Semantics, Idempotence.
 - **Backend & Systems Engineering:** Python (3+ years), PostgreSQL, MySQL, Redis, RESTful Microservices, SaaS API Integrations, OAuth Flows, High-Availability System Design.
 - **RAG & Context Optimization:** Vector Databases, Embedding Models, Strategic Chunking, Hybrid Search, Latency-Optimized Inference Layers.
 - **Observability, Evals & CI/CD:** Signoz, Datadog, Automated Evaluation Frameworks, A/B Testing, Cloud Deployments, Docker, WSL, MCP Servers, Safe Rollbacks.
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Professional Experience

AI/ML Developer

BluePatterns AI | Bengaluru, India | *April 2025 – Present*

- **Autonomous & Self-Healing Agents:** Architected and deployed "aware.ai" an adaptive multi-agent inference layer featuring automated exception fallback logic and self-healing state-recovery loops for complex enterprise tasks.
- **Production-Ready Pipelines:** Built robust, long-running agent execution pipelines using OpenClaw and LangGraph capable of multi-step autonomous reasoning and scheduled background execution.
- **Structured Outputs & Tool Design:** Designed strict, idempotent JSON schemas and tool definitions for agent tool-calling, enabling safe third-party SaaS API

connections and strict enterprise multi-tenancy access controls.

- **Evals & Reliability Frameworks:** Engineered rigorous pre-deployment evaluation pipelines to measure prompt resilience, structural output drift and agent logic, resulting in highly stable production rollouts and safe rollback capabilities.
- **Infrastructure Optimization:** Configured and managed complex local and VPS container environments using Docker and WSL.

Senior System Engineer

Tata Consultancy Services | Bengaluru, India | *December 2022 – May 2025*

- **High-Availability Backends:** Built and maintained enterprise-grade Python microservices, data pipelines and transactional layers using PostgreSQL and Redis under strict SLAs.
- **Secure API Connector Frameworks:** Developed secure third-party integrations, managing OAuth flows, high-concurrency rate limiting and defensive data validation mechanisms across decentralized systems.
- **Production Observability & Recovery:** Managed overall system health and reduced API latency through targeted database indexing and automated query optimization. Actively resolved live production incidents by introducing robust error handling and transactional retry semantics.
- **State & Transaction Integrity:** Spearheaded the refactoring of legacy batch processes to incorporate robust state management, minimizing transactional failures during unexpected downtime.

Engineer Intern

National Institute of Technology, Tiruchirappalli | Tiruchirappalli, India | *2020 – 2021*

- Conducted foundational engineering research and collaborated on technical problem-solving within complex academic and applied frameworks.

Education

- **Master's Degree:** *Amrita Vishwa Vidyapeetham, Coimbatore* | 8.25 CGPA
- **Bachelor's Degree:** *KLE Institute of Technology, Hubli* | 75% FCD VTU

Publications & Certifications

Publications

- *Artist Based Video Recommendation System using Machine Learning.*
- Research Focus: Mitigating information inequality in wide-spread AI dominance,

developing localization frameworks for open-source LLMs, Multi-agentic AI and expanding human-model interaction paradigms.

Certifications

- **Google AI Essentials** | Coursera (*Aug 2024*)
 - **Introduction to Prompt Engineering for Generative AI** | LinkedIn (*Jul 2024*)
 - **Databases and SQL for Data Science with Python (with Honors)** | IBM (*Jan 2023*)
 - **ETL - Extract, Transform and Load Data** | CertNexus (*Jun 2022*)
 - **AWS Machine Learning** | Coursera (*May 2021*)
 - **Machine Learning (Stanford University)** | Coursera (*Apr 2021*)
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Open-Source Contributions

1. Llama3-8b-Wisdom: Fine-tuned LLM & GGUF Release (Hugging Face, GitHub)

- **Optimized Fine-Tuning:** Developed a specialized Llama-3 8B model using Unsloth QLoRA on a curated philosophical dataset. Implemented a custom three-stage prompt architecture (Instruction → Thought → Response) to mitigate conversational drift and ensure precise, contemplative outputs.
- **Quantization & Deployment:** Published multiple GGUF formats (Q4_K_M, Q5_K_M, Q8_0) for efficient CPU-based inference via llama.cpp, providing comprehensive Python quickstart integration paths.
- **Documentation & Governance:** Authored a production-standard model card detailing hyperparameters, evaluation metrics and ethical AI constraints to guide enterprise-level adoption.

2. Qwen3-4B-PromptCraft: Zero-Reasoning Prompt Engineering Model (Hugging Face, GitHub)

- **Deterministic Prompt Generation:** Fine-tuned a Qwen3-4B variant on thousands of synthetic instruction pairs to generate robust system prompts. Engineered the model to suppress internal reasoning tokens, ensuring clean and production-ready outputs.
- **Structured Architecture:** Defined a rigorous three-section output schema for agent workflows, incorporating OWASP LLM security mitigations and comprehensive edge-case handling.
- **Multi-Runtime Availability:** Distributed under Apache 2.0 with optimized GGUF variants and extensive documentation for transformers and llama.cpp runtimes.

3. Aware-ai: Local-First Multi-Agent Inference Router (GitHub)

- **Resilient System Design:** Engineered a dual-plane inference gateway using FastAPI and Ollama, featuring Redis-based state persistence and a comprehensive guardrail suite for PII detection and hallucination filtering.
- **Infrastructure Orchestration:** Developed a modular monorepo with automated agent registration, Docker Compose support and end-to-end validation suites for enterprise environments.
- **Enterprise Privacy Focus:** Configured tunable privacy levels and latency-optimized routing for secure, on-premise AI deployments.

4. **NyayaRakshak:** Sarvam AI-Powered Cyber Crime Complaint Auditor (GitHub, GCP)

- **Legal Process Automation:** Constructed a legal auditor using the Sarvam stack (Aarya, Akshar) to validate complaints against Indian statutory frameworks, including the Bharatiya Nyaya Sanhita 2023.
- **Deployment & Extraction:** Scaled a multi-stage agent pipeline on Google Cloud Run to automate OCR extraction, transactional integrity checks and procedural defect reporting for legal filings.

5. **CyberGuard 1930:** Voice-First Financial Fraud Reporting Portal (GitHub)

- **Multi-Lingual Voice Interface:** Developed a TypeScript-based fraud portal utilizing Sarvam AI (Saaras, Bulbul) to facilitate real-time reporting in Indic languages, reducing victim response time significantly.
- **AI-Driven Risk Scoring:** Implemented an automated entity extraction engine to analyze high-risk fraud patterns, including APK scams and digital extortion, with integrated evidence hash validation.
- **Accessibility & Security:** Built inclusive UI features for senior citizens and secure role-based access for law enforcement, released under an Apache-2.0 license.

6. **EduTech AI Tutor:** Self-Driven Learning Portal (GitHub)

- **Personalised AI Tutoring:** Role-based portals for students, teachers and parents. Students chat with an AI tutor grounded in real NCERT textbooks, request educational diagrams and watch curated video lessons.
- Teachers monitor class engagement and flag struggling students automatically. Parents track their child's progress in real time.

7. **Universal AI Switchboard:** Fundamental breakdown of modern generative AI systems (GitHub)

- **Graph Engineering:** System errors are caught as state updates and routed into self-healing repair loops rather than hard system crashes.
- **Everything is Swappable:** "Lego Bricks" connected over MCP. Swapping an underlying model or database requires changing a single configuration parameter in a `universal_switchboard.yaml` file—zero application code rewrites required.

- **Security & Governance gates:** least-privilege token scoping and mandatory Human-in-the-Loop (HitL) approval checkpoints before high-stakes infrastructure mutations can execute over the network.

8. **QA Agent:** Autonomous QA Agent with Self-Healing Test Automation (GitHub)

- **Orchestration meta-agent:** that accepts a web application URL as its sole required input and executes the full testing lifecycle planning, generation, execution, and repair without human intervention between stages.
 - **Orchestration engine:** is an event-driven deterministic state graph coordinating three decoupled sub-agents (Planner, Generator, Healer) with strict transition guardrails, coverage evaluation gates, and structured schema parsing.
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